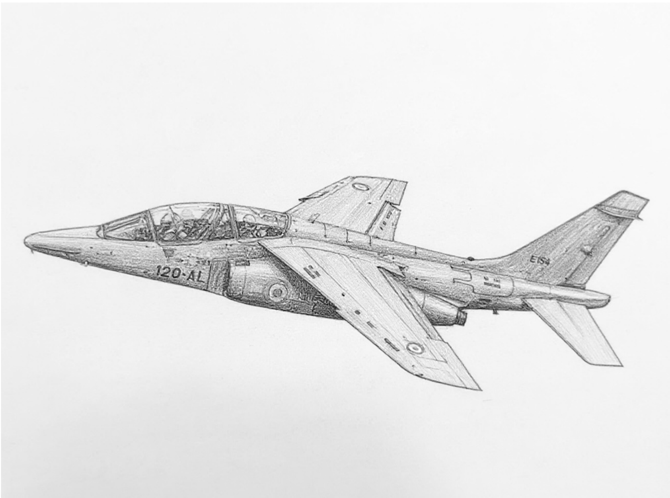
Dassault/Dornier Alpha Jet



The Dassault/Dornier Alpha Jet is a two-seat advanced trainer and light attack aircraft. It was developed starting in 1967 and entered service in 1979.[1]

Versions

Alpha Jet E

The Alpha Jet E is the trainer version, with the E designation taken from *École* (school). It is equipped with two underwing and one centerline weapon stations, but in French and Belgian service these were intended only for weapons training. Some of its later users employed it as a light strike aircraft, despite its shortcomings compared to aircraft dedicated to this role.[2]

It initially served in the French Armée de l'air and the Belgian Air Force, but later with the air forces of Egypt, Morocco, Nigeria, Qatar, Togo, Côte d'Ivoire, and with private companies.[3]

Alpha Jet A

The Alpha Jet A is the attack version, with the A designation taken from *Appui Tactique* (tactical support). Although very similar to the E, the A has two additional underwing weapon stations, a more advanced weapon system, and can replace the rear crew member with an ECM suite.[4]

It initially served in the German Luftwaffe, replacing the Fiat G91R/3. After the end of the Cold War many of these aircraft were sold to the air forces of Portugal and Thailand.[5]

Armament and Stores

The Alpha Jet has no internal guns, but when armed would typically carry a gun pod on the centerline station, normally a CC420 30 mm DEFA pod for E models or a 27 mm Mauser pod for A models. The gun pod was originally not jettisonable, but this restriction was lifted by the 1989 upgrade of Luftwaffe Alpha Jet As.[6]

Typical stores would be SNEB 68 mm or CRV7 70 mm RPs or HE or cluster bombs.[7]

Combat

The Alpha Jet A does not seem to have seen combat, but the Alpha Jet E has in Moroccan and Nigerian service, in the former case against Polisario Front in the Western Sahara War and in the latter against the Boko Haram militant group.

ADCs

– Alpha Jet A
– Alpha Jet E

Notes

1. **Sources.**
   - Goebel, Air Vectors
   - Wikipedia on the Alpha Jet
   - Wikipedia on the DEFA cannon
   - Wikipedia on Gun Pods

   Photo Credit.
   - Tim Felce (CC CC BY 2.5)
2. **Alpha Jet E ADC.** *Air Strike* has an ADC for the E (as a variant of the A). The ADC here is based on Chris Perleberg's update incorporating the changes in APJ 21.
3. **Alpha Jet E Service and Combat.** Wikipedia.
4. **Alpha Jet A ADC.** *Air Strike* has an ADC for the E. The ADC here is based on Chris Perleberg's update incorporating the changes in APJ 21.
5. **Alpha Jet A Service.** Wikipedia.
6. **Gun Pods.**
   - For the A: Single Mauser 27 mm GP (Goebel) with 150 rounds (Wikipedia). The rate of fire is 1000-1700 RPM, which corresponds to 4.5 to 2.6 ammo points. An upgrade in 1989 which Luftwaffe As the ability to jettison the gun pod (Wikipedia).
   - For the E: Single DEFA 30 mm GP (Goebel) with 150 rounds (Wikipedia). The model is the DEFA 553 (Wikipedia on DEFA). The rate of fire is 1300 RPM, which corresponds to 3.5 ammo. It may have been a CC420 gun pod (Wikipedia on gun pods).
7. **Stores.**

   Stores include Matra 18 × 68 mm or CRV7 19 × 70 mm RPs, AIM-9s, Matra Magics, and Mavericks, bombs including the BL755 cluster bomb (Wikipedia). However, the Maverick capability seems to not have been implemented.

   Could carry FTs with capacities of 310L (82 US gal) or 450L (119 US gal) (Goebel).

   The ICE upgrade to the A was to provide compatibility with Mavericks, but this was cancelled in 1988 and a more austere upgrade from 1989 gave the AIM-9Ls and the ability to jettison the gun pod (Wikipedia).

   Goebel states: "The French qualified a film-camera pod".

Alpha Jet A

Crew: Pilot and Observer

Power APs/DPs: ○○

	CL	1/2	DT	Fuel
AB	—	—	—	—
M	1.5	1.0	1.0	2.0
N	0.0	0.0	0.0	1.0
I	1.0	1.0	1.0	0.0
SPBR	1.0	2.0	2.0	—

Maneuver HFPs/DPs:

LR/DR	1.0	1.0
VR		0.0

Turn DPs:

	CL	1/2	DT
TT	1.0	1.0	1.0
HT	1.0	2.0	2.0
BT	3.0	3.0	4.0
ET	—	—	—

Cruise Spd. CL:	5.0	Restr. Arcs:	—
Climb Spd.:	4.0	Blind Arcs:	30–
Visibility:	4	Internal Fuel:	180
Size:	+1	AtA Refuel:	No
Vulnerability:	−1	Ejection Seat:	Std

Speeds and Ceilings						Climb Capabilities						
Alt. Band	Conf. Ceil.	CL 45	1/2 38	DT 32	Dive Speed	CL AB	CL Oth	1/2 AB	1/2 Oth	DT AB	DT Oth	
EH+	46+	—	—	—	—	—	—	—	—	—	—	EH+
VH	36–45	2.5 – 5.5	3.0 – 5.0	—	7.0	—	0.5	—	0.5	—	—	VH
HI	26–35	2.5 – 6.0	2.5 – 5.5	2.5 – 5.0	7.0	—	1.0	—	0.5	—	0.5	HI
MH	17–25	2.0 – 6.0	2.0 – 5.5	2.5 – 5.0	7.0	—	1.0	—	1.0	—	0.5	MH
ML	8–16	1.5 – 6.0	2.0 – 5.5	2.0 – 5.0	7.0	—	1.0	—	1.0	—	1.0	ML
LO	0–7	1.0 – 6.0	1.5 – 5.5	1.5 – 5.0	7.0	—	2.0	—	1.5	—	1.0	LO

Radar:	—
ECCM:	—
Arcs:	—
Search:	—
Track:	—
Lock-On:	—

ECM:	IFF
RWR:	A
DDS:	—
DJM:	B3
AJM:	—
BJM:	—

Weapon Stations Diagram:

Guns:	—
To Hit:	—
Ammunition:	—
Gunsight:	TT+0/HT+1/BT+2
Ranging:	—
AtA/AtG:	—

Bomb System: Ballistic

Technology:

None

Load Point Limits: CL : 0–4, 1/2: 5–8, DT : 9+

Weight Limit: 5,550

Station	Limit	Allowed Loads
1 and 5	750	FT BB BG RP IRM EP
2 and 4	1,350	BB BG RP DR EP
3	1,350	BB BG RP GP PP

Load Notes:
1. May use 310L and 450L FTs
2. May use 27 mm Mauser BK-27 or CC420 30 mm DEFA GPs. Prior to 1989, the GP cannot be jettisoned.
3. From 1989, may use AIM-9 and Matra 550 IRMs.

Notes:
1. The Dassault/Dornier Alpha Jet A is a light attack aircraft.
2. High transonic drag (HTD).
3. Either the observer or the RWR and DJM may be carried.

VPs: 12/8/4/2

Alpha Jet E

Crew: Pilot and Observer

Power APs/DPs: ○○

	CL	1/2	DT	Fuel
AB	—	—	—	—
M	1.5	1.0	1.0	2.0
N	0.0	0.0	0.0	1.0
I	1.0	1.0	1.0	0.0
SPBR	1.0	2.0	2.0	—

Maneuver HFPs/DPs:

LR/DR	1.0	1.0
VR		0.0

Turn DPs:

	CL	1/2	DT
TT	1.0	1.0	1.0
HT	1.0	2.0	2.0
BT	3.0	3.0	4.0
ET	—	—	—

Cruise Spd. CL:	5.0	Restr. Arcs:	—
Climb Spd.:	4.0	Blind Arcs:	30–
Visibility:	4	Internal Fuel:	180
Size:	+1	AtA Refuel:	No
Vulnerability:	−1	Ejection Seat:	Std

Speeds and Ceilings						**Climb Capabilities**						
Alt. Band	Conf. Ceil.	CL 45	1/2 38	DT 32	Dive Speed	CL AB	CL Oth	1/2 AB	1/2 Oth	DT AB	DT Oth	
EH+	46+	—	—	—	—	—	—	—	—	—	—	EH+
VH	36–45	2.5 – 5.5	3.0 – 5.0	—	7.0	—	0.5	—	0.5	—	—	VH
HI	26–35	2.5 – 6.0	2.5 – 5.5	2.5 – 5.0	7.0	—	1.0	—	0.5	—	0.5	HI
MH	17–25	2.0 – 6.0	2.0 – 5.5	2.5 – 5.0	7.0	—	1.0	—	1.0	—	0.5	MH
ML	8–16	1.5 – 6.0	2.0 – 5.5	2.0 – 5.0	7.0	—	1.0	—	1.0	—	1.0	ML
LO	0–7	1.0 – 6.0	1.5 – 5.5	1.5 – 5.0	7.0	—	2.0	—	1.5	—	1.0	LO

Radar:	—
ECCM:	—
Arcs:	—
Search:	—
Track:	—
Lock-On:	—

ECM:	IFF
RWR:	—
DDS:	—
DJM:	—
AJM:	—
BJM:	—

Weapon Stations Diagram:

Guns:	—
To Hit:	—
Ammunition:	—
Gunsight:	TT+0/HT+1/BT+2
Ranging:	—
AtA/AtG:	—

Bomb System: Manual

Technology:

None

Load Point Limits: CL : 0–4 · 1/2: 5–8 · DT : 9+

Weight Limit: 5,550

Station	Limit	Allowed Loads
1 and 3	750	FT BB BG RP IRM EP
2	1,350	BB BG RP GP PP

Load Notes:
1. May use 310L and 450L FTs
2. May use 27 mm Mauser BK-27 or CC420 30 mm DEFA GPs. Prior to 1989, the GP cannot be jettisoned.
3. May use AIM-9 and Matra 550 IRMs.

Notes:
1. The Dassault/Dornier Alpha Jet E is a trainer with a secondary light attack capability.
2. High transonic drag (HTD).

VPs: 10/7/3/2